Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 14: Approved Solutions

Double Inclusion, Disproof, and Partitions

Covers Chartrand & Zhang, Sections 2.2 and 2.4

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first — and for the proof problems, write in *sentences*. That is the skill being practised.

Part I: How to Practice This Material

The template, and the two halves

Proving $A = B$

Half one. Let $x \in A$. . . . reasoning . . . Therefore $x \in B$. Hence $A \subseteq B$.
Half two. Let $y \in B$. . . . reasoning . . . Therefore $y \in A$. Hence $B \subseteq A$.
Conclude. Since each is a subset of the other, $A = B$.
The element is *arbitrary*: you assume only that it belongs to the set you started with. That is what makes the conclusion cover every element at once.

Disproving is much cheaper

One specific pair of sets, and one specific element in one side but not the other. Name both. "It doesn't always work" is not a disproof.

Three mistakes to expect

1. **Proving one inclusion and stopping.** Set equality is a biconditional.
2. **Assuming something about x.** If you use a property the arbitrary element need not have, the argument covers only some elements.
3. **Writing symbols with no sentences.** A proof is prose. Chains of implications with no words are not gradeable.

Part II: Practice Problems

Section A — Proving

1. Prove that $A \cap (B - C) = (A \cap B) - C$ for all sets A, B, C.

Approved solution

Half one: $A \cap (B - C) \subseteq (A \cap B) - C$.
Let $x \in A \cap (B - C)$. Then $x \in A$ and $x \in B - C$. From the second, $x \in B$ and $x \notin C$. Since $x \in A$ and $x \in B$, we have $x \in A \cap B$. Together with $x \notin C$, this gives $x \in (A \cap B) - C$.
Half two: $(A \cap B) - C \subseteq A \cap (B - C)$.
Let $y \in (A \cap B) - C$. Then $y \in A \cap B$ and $y \notin C$. From the first, $y \in A$ and $y \in B$. Since $y \in B$ and $y \notin C$, we have $y \in B - C$. Together with $y \in A$, this gives $y \in A \cap (B - C)$.
Since each side is a subset of the other, the two sets are equal. ■

Note how symmetric the halves are. Both amount to observing that "$x \in A$ and $x \in B$ and $x \notin C$" can be grouped either way — which is the associative law for $\wedge$. When a proof's two halves read almost identically, that is usually why.

2. Prove that $A - (A - B) = A \cap B$ for all sets A and B.

Approved solution

Half one: $A - (A - B) \subseteq A \cap B$.
Let $x \in A - (A - B)$. Then $x \in A$ and $x \notin A - B$. Now $x \notin A - B$ means it is *not* the case that ($x \in A$ and $x \notin B$). Since we know $x \in A$, the failure must be in the second part, so $x \in B$. Hence $x \in A$ and $x \in B$, giving $x \in A \cap B$.
Half two: $A \cap B \subseteq A - (A - B)$.
Let $y \in A \cap B$, so $y \in A$ and $y \in B$. Since $y \in B$, y does not satisfy "$\in A$ and $\notin B$," so $y \notin A - B$. Combined with $y \in A$, this gives $y \in A - (A - B)$.
Therefore $A - (A - B) = A \cap B$. ■

Half one uses De Morgan. "Not ($x \in A$ and $x \notin B$)" becomes "$x \notin A$ or $x \in B$," and the first disjunct is ruled out by what we already know. Spelling that step out is what separates a proof from a plausible paragraph.

Section B — Disproving

3. Each claim below is *false*. Give a counterexample for each: specific sets, and the specific element that fails.

(a) For all sets A and B, $\quad A - B = B - A$.

(b) For all sets A, B, C, $\quad A - (B - C) = (A - B) - C$.

Approved solution

(a) Take $A = \{1, 2\}$ and $B = \{2, 3\}$. Then $A - B = \{1\}$ and $B - A = \{3\}$. The element 1 is in the left side and not the right, so the sets differ.
(b) Take $A = \{1\}$, $B = \{1\}$, $C = \{1\}$. Then $B - C = \emptyset$, so $A - (B - C) = A = \{1\}$. But $A - B = \emptyset$, so $(A - B) - C = \emptyset$. The element 1 is in the left side and not the right.

Part (b) is the instructive one: set difference is *not associative*. Students who expect it to behave

like subtraction of numbers are also wrong there — $a-(b-c) \neq (a-b)-c$ in arithmetic too. The analogy holds here; it is the parentheses that matter.

4. Determine whether $(A \cup B) - B = A$ holds for all sets. Prove it or disprove it. Then state and justify what the left side *does* equal.

Approved solution

False. Take $A = \{1, 2\}$ and $B = \{2\}$. Then $A \cup B = \{1, 2\}$, so $(A \cup B) - B = \{1\}$, while $A = \{1, 2\}$. The element 2 is in A and not in the left side.
What it does equal: $(A \cup B) - B = A - B$.
Proof. Let $x \in (A \cup B) - B$. Then $x \in A \cup B$ and $x \notin B$. Since $x \in A$ or $x \in B$, and the second is ruled out, $x \in A$. Hence $x \in A - B$.
Conversely, let $y \in A - B$, so $y \in A$ and $y \notin B$. Then $y \in A \cup B$, and $y \notin B$, so $y \in (A \cup B) - B$. ■

The claim fails exactly on $A \cap B$ — the part of A that B removes. Identifying *where* a false claim goes wrong is worth more than the counterexample by itself, and it is the difference between a fix and a shrug.

Section C — Partitions

5. Let $A = \{1, 2, \ldots, 8\}$. Which of these are partitions of A? For those that are not, say exactly which condition fails.

(a) $\{1, 3, 5, 7\}$, $\{2, 4, 6, 8\}$

(b) $\{1, 2, 3\}$, $\{4, 5\}$, $\{5, 6, 7, 8\}$

(c) $\{1, 2\}$, $\{3, 4\}$, $\{5, 6\}$, $\{7\}$

(d) $\{1, 2, 3, 4\}$, $\emptyset$, $\{5, 6, 7, 8\}$

Approved solution

(a) Yes. Odds and evens: disjoint, nonempty, union is A.
(b) No. The element 5 belongs to two blocks, so they are not pairwise disjoint. Every element must be in *exactly* one.
(c) No. The element 8 belongs to no block, so the union is not A.
(d) No. One block is empty, and partition blocks must be nonempty.

Three ways to fail — overlap, omission, and an empty block — and this problem has one of each. When checking a candidate partition, check all three.

Section D — Writing

6. In five to eight complete sentences, explain why proving $A = B$ requires two separate arguments while disproving it requires only one example. Connect your answer to the definition of set equality and to something you learned before this week.

Approved solution

Model answer. By definition, $A = B$ means $A \subseteq B$ and $B \subseteq A$, and a conjunction is true only when both parts are, so establishing equality requires establishing both inclusions. Each inclusion is itself a universal claim — $A \subseteq B$ says every element of A lies in B — which is why each half is proved by taking an arbitrary element and reasoning about it rather than by checking examples. Disproving $A = B$ is the negation of that conjunction, and by De Morgan the negation of "both hold" is "at least one fails," so a single failing inclusion is enough. To exhibit a failing inclusion you need only one element in one set and not the other. This is the same asymmetry as universal and existential statements from last week: proving something about all elements takes an argument, and refuting it takes a witness.

Full credit requires the connection to the conjunction in the definition, and to the universal-versus-existential asymmetry. An answer that only describes the template earns about half.

Section E — Challenge (optional)

7. Prove that if $A \subseteq B$, then $A \cup B = B$. Then state and prove the converse.

Approved solution

Forward. Suppose $A \subseteq B$. To show $A \cup B = B$, prove both inclusions.
$B \subseteq A \cup B$ holds for any sets, since $y \in B$ gives $y \in A \cup B$ immediately.
For $A \cup B \subseteq B$: let $x \in A \cup B$, so $x \in A$ or $x \in B$. If $x \in A$ then $x \in B$ because $A \subseteq B$; if $x \in B$ we are already done. Either way $x \in B$.
Hence $A \cup B = B$.
Converse. *If $A \cup B = B$, then $A \subseteq B$.* Suppose $A \cup B = B$ and let $x \in A$. Then $x \in A \cup B$, and since $A \cup B = B$, we get $x \in B$. Hence $A \subseteq B$.

Both directions hold, so $A \subseteq B$ *if and only if* $A \cup B = B$. That is a biconditional, and as Lecture 6 said, it needed two proofs.
The forward direction used a case split on $x \in A$ or $x \in B$ — the standard move when a hypothesis is a disjunction. You will use it constantly in Chapter 3.

Instructor notes

- **These are the first real proofs students write.** Grade the prose as hard as the logic. A correct chain of symbols with no sentences should not earn full marks now, or the standard will not hold in Chapter 3.
- **Problem 2, half one** is where De Morgan does hidden work. Students often assert "so $x \in B$" without saying why. Make them write the step.
- **Problem 4** asks not just for a disproof but for the correct statement. That second half is the more valuable habit and is worth most of the credit.
- **Problem 5** has one of each failure mode — overlap, omission, empty block. Useful as a checklist when partitions return with equivalence relations.
- **Problem 7** introduces the case split on a disjunctive hypothesis. Name the technique aloud; it is the workhorse of direct proof.
- Reserved for Homework 6: even-numbered exercises from Sections 2.2 and 2.4.